

CODE_DDP: Pooling-aware Final-token Adapter Bank on Prompted 197 Tokens

Task-native prompted-token training · point-wise adaptation of all 197 tokens · pooling + attention + margin preservation · exact DDP pooling

Text branch

Negative DDP prompt

Positive DDP prompt

Frozen text encoder · DDP text feature

$$\mathbf{P}_T^{c-}, \quad \mathbf{P}_T^{c+}, \quad \mathbf{t}^{c\pm} = \text{norm}(\mathbf{E}_T(\mathbf{P}_T^{c\pm}, c))$$

Frozen prompted-text reference shared by teacher and adapted route

$$\mathbf{t}_{\text{ref}}^{c\pm} = \text{sg}(\mathbf{t}^{c\pm}), \quad \hat{\mathbf{t}}^{c\pm} = \mathbf{t}^{c\pm} \quad (\text{frozen text side})$$

Image branch

Image input

Class-specific dual visual prompts

Frozen visual encoder with DDP prompts

$$\mathbf{x}$$

$$\mathbf{P}_V^c = \{\mathbf{P}_V^{c-}, \mathbf{P}_V^{c+}\}$$

$$\mathbf{F}^{c\pm} = \mathbf{E}_V(\mathbf{P}_V^{c\pm}, \mathbf{x}) \in \mathbb{R}^{(N+1) \times d}$$

Full projected token sequence

$$\mathbf{F}^{c\pm} = [\mathbf{f}_0^{c\pm}, \dots, \mathbf{f}_N^{c\pm}], \quad N = 196$$

Prompted CLS representation

$$\mathbf{f}_{\text{cls}}^{c\pm} = \mathbf{f}_0^{c\pm}$$

$$\mathbf{F}_{\text{ref}}^{c\pm} = \text{sg}(\mathbf{F}^{c\pm}), \quad \mathbf{F}_{\text{student}}^{c\pm} = \mathbf{F}^{c\pm}, \quad N + 1 = 197$$

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Explicit feature ports from the shared text/image branches into two routes

Prompted tokens / DDP text → frozen Task-routed Adapter Bank → original DDP pooling

$$\mathbf{F}^{c\pm}$$

$$\mathbf{t}^{c\pm}$$

Inference route: class-routed Pooling-aware Adapter on all 197 tokens → recomputed DDP pooling

Frozen Adapter cards accumulated over tasks · route each class to its introduction-task Adapter

$$A_0[\mathcal{C}^0]$$

$$A_1[\mathcal{C}^1]$$

$$\dots$$

$$A_i[\mathcal{C}^i]$$

Deterministic class routing and frozen Adapter application before DDP

$$\begin{aligned} \tau(c) &= s \iff c \in \mathcal{C}^s, \quad n = 0, \dots, N, \\ \mathbf{r}_{n, \tau(c)}^{c\pm} &= W_{2, \tau(c)} \sigma(W_{1, \tau(c)} \text{norm}(\mathbf{f}_n^{c\pm})), \\ \hat{\mathbf{f}}_n^{c\pm} &= \text{norm}(\mathbf{f}_n^{c\pm} + \alpha \mathbf{r}_{n, \tau(c)}^{c\pm}), \quad \alpha = 0.03 \end{aligned}$$

Recompute the exact original DDP token-text logits

$$\hat{q}_n^{c\pm} = 20(\mathbf{t}^{c\pm})^\top \hat{\mathbf{f}}_n^{c\pm}$$

Positive-path attention shared by the matching negative path

$$\tilde{\omega}_n^c = \frac{\exp(\hat{q}_n^{c+})}{\sum_{m=0}^N \exp(\hat{q}_m^{c+})}$$

Original shared-attention pooling over the adapted token sequence

$$\tilde{\mathbf{f}}^{c\pm} = \sum_{n=0}^N \tilde{\omega}_n^c \hat{\mathbf{f}}_n^{c\pm}$$

Original DDP positive / negative path-logit head

$$\tilde{\ell}^{c\pm} = 5 \sum_{n=0}^N \tilde{\omega}_n^c \hat{q}_n^{c\pm} = 100(\mathbf{t}^{c\pm})^\top \tilde{\mathbf{f}}^{c\pm}$$

Final temperature-scaled two-way prediction

$$\hat{\mathbf{y}}^t = [\hat{y}_{c+}^t]_{c \in \mathcal{C}^{0:t}}, \quad \hat{y}_{c+}^t = \frac{\exp(\tilde{\ell}^{c+}/\tau(t))}{\exp(\tilde{\ell}^{c+}/\tau(t)) + \exp(\tilde{\ell}^{c-}/\tau(t))}$$

Prompted 197 tokens / DDP text → Pooling-aware Final-token Adapter training

Pooling-aware Final-token Adapter training route

TRAINING ONLY

Current-task prompted-token cache and strict partial-label mask

$$\begin{aligned} \mathcal{D}_s^{\text{Full}} &= \{(\mathbf{F}_i, \mathbf{t}, \mathbf{y}_i) : \sum_{c \in \mathcal{C}^s} y_{ic} > 0\}, \\ M_{ic}^s &= 1[c \in \mathcal{C}^s] \end{aligned}$$

Train the new task Adapter point-wise on all 197 prompted final tokens

$$\begin{aligned} \mathbf{r}_{i,n,s}^{c\pm} &= W_{2,s} \sigma(W_{1,s} \text{norm}(\mathbf{f}_{i,n}^{c\pm})), \\ \hat{\mathbf{f}}_{i,n,s}^{c\pm} &= \text{norm}(\mathbf{f}_{i,n}^{c\pm} + 0.03 \mathbf{r}_{i,n,s}^{c\pm}), \quad n = 0, \dots, 196 \end{aligned}$$

Frozen original DDP teacher and adapted-token student quantities

$$\begin{aligned} (\omega_i^c, \tilde{\mathbf{f}}_i^{c\pm}, \ell_i^{c\pm}, m_i^c) &= \text{DDP}(\mathbf{F}_i^{c\pm}, \mathbf{t}^{c\pm}), \\ (\tilde{\omega}_i^c, \tilde{\mathbf{f}}_i^{c\pm}, \tilde{\ell}_i^{c\pm}, \tilde{m}_i^c) &= \text{DDP}(\tilde{\mathbf{F}}_{i,s}^{c\pm}, \mathbf{t}^{c\pm}), \quad m_i^c = \ell_i^{c+} - \ell_i^{c-} \end{aligned}$$

Masked classification with token, pooling, attention and margin preservation

$$\begin{aligned} \mathcal{L}_s &= \text{BCE}_{M^s}(\tilde{\mathbf{m}}, \mathbf{y}) + 0.1 \mathcal{L}_{\text{token}} + 100 \mathcal{L}_{\text{pool}} + 100 \mathcal{L}_{\text{attn}} + \mathcal{L}_{\text{margin}}, \\ \mathcal{L}_{\text{pool}} &= 1 - \cos(\tilde{\mathbf{f}}, \tilde{\mathbf{f}}), \quad \mathcal{L}_{\text{attn}} = D_{\text{KL}}(\omega \parallel \tilde{\omega}), \quad \mathcal{L}_{\text{margin}} = \text{SmoothL1}(\tilde{\mathbf{m}} - \mathbf{m}) \end{aligned}$$

Freeze the learned task mapping and insert one new Adapter card

$$(W_{1,s}^*, W_{2,s}^*) = \arg \min \mathcal{L}_s, \quad A_s \xrightarrow{\text{freeze}} \{A_0, \dots, A_s\}$$

Strict incremental constraints and fixed inference protocol

$$A_{0:s-1} \text{ frozen; } \mathbf{y}^{\text{old/future}} \text{ masked; } \alpha = 0.03; \quad \mathcal{D}_{\text{test}} \text{ unseen}$$

shared text representations

shared image representations

frozen class-routed Adapter Bank

original DDP pooling after adaptation

training-only prompted pooling-aware route